# Aeronautics and Space Technology Club

## IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 13 · Part D: Digital Signal Processing Level 3

**Track:** Periodogram, Welch, Multitaper, Lomb-Scargle Periodogram, Period Detection

**Prerequisites:** Problems 07, 12 · **Estimated effort:** 12–16 hours

**Spectral Estimation: From a Single Noisy Record to a Trustworthy Spectrum**

The periodogram of a random signal is inconsistent: more data does not make it less noisy. Measure that scatter, fix it with Welch averaging and multitaper estimation, attach correct confidence intervals, read power-law slopes of coloured noise, and find the period of a pulsating star from unevenly sampled data with the Lomb–Scargle periodogram.

## General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

| Stage | Phase | What you should have at the end of it |
|---|---|---|
| 1 | Think (∼20%) | Theory understood; every analytic result derived; signals, units and sampling rates written down. |
| 2 | Build (∼30%) | Core code working end to end on the baseline data and validated against the analytic results. |
| 3 | Experiment (∼30%) | Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers. |
| 4 | Refine (∼20%) | Weaknesses fixed, honest analysis and memo written, repository folder clean. |

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/13_spectral_estimation/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of $2\pi$, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

## Background and Motivation

Problem 12 showed what the DFT does to a deterministic signal. For noise and for signals that are only *statistically* periodic, a different question arises: given one finite record, what is the underlying power spectral density, and how accurate is the estimate? The raw periodogram is an unbiased-looking but statistically useless estimate, because each frequency bin is a $\chi^2$ variable with just two degrees of freedom no matter how long the record. Averaging trades frequency resolution for variance; multitaper methods do so more efficiently.

Astronomical and flight data add a complication: observations are rarely evenly spaced. A star is observed on clear nights; telemetry has dropouts. The FFT assumes uniform sampling, so the Lomb–Scargle periodogram is the standard tool, and its false-alarm probability is how a period is claimed as a detection.

## Theoretical Framework

### 1. Periodogram

For a length-$N$ record with window $w$,

$$\hat{S}_{\text{per}}[k] = \frac{|X_w[k]|^2}{f_s \sum_n w[n]^2} \text{ (two-sided).} \tag{1}$$

Its expectation is the true PSD convolved with the window's spectral kernel (leakage bias), and its variance is $\text{Var}\{\hat{S}\} \approx S^2$ for Gaussian noise, independent of $N$: each bin is $S\,\chi^2_2/2$.

### 2. Bartlett and Welch

Splitting the record into $K$ segments of length $L$, windowing and averaging the periodograms reduces the variance by about $1/K$. Welch's method uses overlapping segments (50% for Hann) for a better compromise; the resolution is set by the segment length, $\Delta f \approx \text{ENBW} \cdot f_s/L$. With $\nu$ equivalent degrees of freedom ($\nu \approx 2K$ for non-overlapping segments), a $(1-\alpha)$ confidence interval for the true PSD is

$$\left[\frac{\nu\,\hat{S}}{\chi^2_{\nu,\,1-\alpha/2}},\ \frac{\nu\,\hat{S}}{\chi^2_{\nu,\,\alpha/2}}\right]. \tag{2}$$

For $K = 10$ non-overlapping segments the 95% interval is $[0.585,\ 2.085]\,\hat{S}$.

### 3. Multitaper

The Thomson multitaper method averages the periodograms obtained with $K = 2NW - 1$ orthogonal discrete prolate spheroidal (DPSS) tapers of time-half-bandwidth product $NW$; it has low leakage (important for spectra with large dynamic range) and variance reduced by about $1/K$ without segmenting the record.

### 4. Coloured Noise

A process with $S(f) \propto f^{-\alpha}$ appears as a straight line of slope $-\alpha$ on a log–log plot. Steep spectra suffer from periodogram leakage, which flattens the high-frequency end unless a window with low side-lobes is used.

### 5. Lomb–Scargle

For observations $y_i$ at arbitrary times $t_i$ (mean subtracted), the Lomb–Scargle power at angular frequency $\omega$ is

$$P(\omega) = \frac{1}{2}\left\{ \frac{\left[\sum_i y_i \cos\omega(t_i-\tau)\right]^2}{\sum_i \cos^2\omega(t_i-\tau)} + \frac{\left[\sum_i y_i \sin\omega(t_i-\tau)\right]^2}{\sum_i \sin^2\omega(t_i-\tau)} \right\}, \quad \tan 2\omega\tau = \frac{\sum_i \sin 2\omega t_i}{\sum_i \cos 2\omega t_i}. \tag{3}$$

It is equivalent to least-squares fitting of a sinusoid at each frequency. For pure Gaussian noise of variance $\sigma^2$ the normalised power $z = P(\omega)/\sigma^2$ is exponentially distributed with unit mean ($\Pr\{z > z_0\} = \mathrm{e}^{-z_0}$), so the false-alarm probability of the highest peak among $N_f$ independent frequencies is $\mathrm{FAP} \approx 1-(1-\mathrm{e}^{-z})^{N_f}$. Uneven sampling imprints a *spectral window* on the periodogram: sampling once per night produces aliases at $f_0 \pm k\,1/\mathrm{d}$.

## Data and Simulation Specification

| | |
|---|---|
| **White and coloured noise** | $f_s = 1000\,\mathrm{Hz}$, $N = 2^{12}, \ldots, 2^{18}$; AR(4) process with a spectral dynamic range of about 80 dB (generate with a pole-zero design of your choice); $1/f^{\alpha}$ noise with $\alpha = 1, 2$. |
| **Line plus noise** | Sinusoid at 123.4 Hz, amplitude 0.05, in white noise $\sigma = 1$, $N = 2^{16}$. |
| **Uneven star** | `sp_common.synth.uneven_variable_star(n=400, period=0.6, amp=0.25, noise=0.04, baseline_days=800)`: nightly sampling with a seasonal gap. |
| **Light curve** | `sp_common.synth.transit_lightcurve()` for the spectral window and detection of stellar variability near 11 and 5.3 d. |
| **Real data (choose one)** | Monthly mean sunspot number from SILSO (find the $\approx$ 11-year cycle); AAVSO photometry of a pulsating variable star; a TESS light curve through `lightkurve`. |

## Your Task

1. **Periodogram statistics (Stage 2).** For white noise with $N = 2^8, \ldots, 2^{16}$ show that the scatter of the periodogram bins does not decrease with $N$, that $\hat{S}/S$ is $\chi^2_2/2$ distributed (histogram and Kolmogorov–Smirnov test) and that the relative standard deviation is 1.
2. **Welch from scratch.** Implement Bartlett and Welch estimators with window, overlap and detrending options. Verify against `scipy.signal.welch`. Plot the variance reduction against $K$ and the resolution loss, and add confidence intervals from Eq. (2); check their empirical coverage over 1000 realisations. Choose the segment length for the line-plus-noise signal and justify it.

3. **Leakage and dynamic range.** Estimate the high-dynamic-range AR(4) spectrum with the rectangular, Hann and Blackman–Harris windows and with Welch's method; show the leakage floor of each against the true PSD computed from the model.
4. **Multitaper.** Implement the multitaper estimator with `scipy.signal.windows.dpss` for $NW = 2, 3, 4$. Compare bias and variance with Welch on the AR(4) spectrum and on the line-plus-noise signal; discuss the cost in resolution.
5. **Spectral slopes.** Estimate the exponent of the $1/f$ and $1/f^2$ noises on a log–log plot with a line fit and its uncertainty. Show the bias of the periodogram for $\alpha = 2$ and its cure.
6. **Lomb–Scargle.** Implement Eq. (3) directly and check against `scipy.signal.lombscargle`. Find the period of the synthetic star, display the spectral window, identify the one-per-day aliases, compute the false-alarm probability by bootstrap (shuffle the magnitudes, 1000 times) and compare with the analytic formula. Explain why the second harmonic of the star's non-sinusoidal light curve also appears.
7. **Real data.** Apply the appropriate estimator to your real series (Welch for the evenly spaced sunspot data, Lomb–Scargle for the uneven photometry). Report the period with an uncertainty from the peak width or a bootstrap and test the peak against the red-noise alternative.

**Self-check targets**

For $K = 10$ non-overlapping averages the 95% confidence factors are 0.585 and 2.085. The synthetic star has $f_0 = 1/0.6 = 1.667$ cycles per day, with daily aliases at 0.667 and 2.667 cycles per day; the strongest peak must lie at $1.667 \pm 0.002$ cycles per day. For white noise of one-sided density $S$ every periodogram bin has relative standard deviation 1.

## Aerospace Application Link

Pointing jitter of a spacecraft telescope is specified as an RMS value over a band, computed by integrating the vibration PSD. From a gyro record (use the synthetic gyro of problem 01 or the machine signal of problem 12) estimate the PSD with Welch and multitaper, compute the cumulative RMS jitter as a function of frequency, and report the RMS in the band 1–50 Hz with its confidence interval. Identify the frequencies that contribute more than 50% of the jitter and what you would change (isolation, wheel speed) to reduce it.

## Deliverables

**D1** **Periodogram statistics:** scatter versus $N$, $\chi^2_2$ test.

**D2** **Welch implementation:** validation against SciPy, variance and resolution trade-off, confidence-interval coverage.

**D3** **Leakage and multitaper:** AR(4) spectral comparison and multitaper bias and variance study.

**D4** **Slopes and Lomb–Scargle:** slope fits with uncertainties, the Lomb–Scargle implementation, spectral window, alias identification and FAP.

**D5** **Real-data period or cycle analysis** with uncertainty and a red-noise significance test.

**D6** **Memo (250 words):** "How do we claim a period is real?" Include the false-alarm argument and cite at least two external sources.

**Bonus** (a) Implement the fast Lomb–Scargle algorithm of Press and Rybicki and benchmark it. (b) Add the floating-mean (generalised) Lomb–Scargle and compare for short records. (c)

Perform spectral pre-whitening and CLEAN to find the second period in the star's signal.

## Learning Resources

### Spectral Estimation

- Stoica, P., Moses, R. *Spectral Analysis of Signals* (periodogram, Welch, Blackman–Tukey, parametric methods).
- Welch, P.D. (1967). "The use of fast Fourier transform for the estimation of power spectra." IEEE Trans. Audio Electroacoust. 15, 70.
- Thomson, D.J. (1982). "Spectrum estimation and harmonic analysis." Proc. IEEE 70, 1055.
- Percival, D.B., Walden, A.T. *Spectral Analysis for Physical Applications.*

### Uneven Sampling

- Lomb, N.R. (1976). "Least-squares frequency analysis of unequally spaced data." Astrophys. Space Sci. 39, 447.
- Scargle, J.D. (1982). "Studies in astronomical time series analysis. II." Astrophys. J. 263, 835.
- VanderPlas, J.T. (2018). "Understanding the Lomb–Scargle periodogram." Astrophys. J. Suppl. 236, 16, arXiv:1703.09824.

### Data and Tools

- SILSO sunspot number: `https://www.sidc.be/SILSO/`; AAVSO data: `https://www.aavso.org`; MAST/TESS with Lightkurve: `https://lightkurve.github.io/lightkurve/`
- SciPy `signal.welch`, `signal.windows.dpss`, `signal.lombscargle`: `https://docs.scipy.org/doc/scipy/reference/signal.html`

## Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

| Criterion | What to look for | Marks |
|---|---|---|
| **Conceptual depth** | Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour. | /20 |
| **Implementation quality** | Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass. | /20 |
| **Aerospace-application integration** | The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires. | /20 |
| **Analysis and critical thinking** | Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases. | /20 |
| **Communication** | Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow. | /20 |

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.